

1(a)(i)	<i>idea that</i> not measured / AW ;	1	e.g. results show colours / no numbers / subjective / observations / description not numbers I categoric
1(b)	<i>2 of:</i> <i>idea that:</i> they are all acting as controls ; <i>idea that:</i> they show that colour change or <u>the effect</u> occurs, if DCPIP chloroplast and light are present ; <i>tube 2:</i> shows that chloroplast suspension does not change colour on its own or without DCPIP there is no colour change or DCPIP is needed for the colour change ; <i>tube 3:</i> shows that DCPIP does not change colour on its own or without chloroplast suspension there is no colour change or chloroplast suspension is needed for the colour change ; <i>tube 4:</i> shows that DCPIP and chloroplast suspension together do not change colour on their own or without light there is no colour change or light is needed for the colour change ;	2	R controlled variable / controlled experiment <i>tube 2:</i> A <i>idea that:</i> acts as a colour comparison for the end point / tube 1
1(c)	<i>idea that:</i> reducing heat (from lamp) passing through / AW ;	1	

Question	Answer	Marks	Guidance
1(d)(i)	<i>independent:</i> light intensity / distance of tube from light source / lamp ; <i>dependent:</i> time taken for DCPIP to decolourise / time for green colour to appear ;	2	I ref. to rate of light dependent reaction I ref to solution / tube, decolourising / going colourless

Question	Answer	Marks	Guidance
1(d)(ii)	7 of: 1. method to vary light intensity ; 2. five values of stated distances in the range 10 to 200 cm ; <i>critical step</i> 3. start timer straight away when exposed to light ; 4. record time, (at each intensity) until DCPIP decolourises / chloroplast suspension is green ; 5. use a colour comparator / tube 2, to decide when to stop timing ; 6. same volume of (same) chloroplast suspension ; 7. same volume of DCPIP ; 8. same concentration of DCPIP ;	7	1. e.g. distances between tube and lamp or different strengths of neutral density filters or wattages of bulbs / variable resistor / dimmer switch / number of lamps 2. for other stated methods, must have 5 stated values, e.g. wattages/ etc. 4. A record time, until blue colour disappears / colour changes R tube goes colourless without ref. to DCPIP if using colorimeter: mp 4 –removes samples at intervals until absorbance becomes zero or constant mp 5 – calibrating / zeroing with chloroplast suspension mp6 & mp7: if volumes are stated, total volume must be minimum 10 cm³ and maximum 50 cm³ (so it fits in the tube) if outside this range award mp 6 or mp 7 but not both I 'same volume' of tube contents

Question	Answer	Marks	Guidance
1(d)(ii)	9. water tank at same distance from test tube / light source ; 10. ref. to a minimum of two replicates <u>and</u> calculate a mean or identify / eliminate/ remove anomalies ; 11. medium risk or hazard <u>and</u> suitable safety ;		
1(e)	2.5×10^{-3} / 0.0025 ;	1	A 25 if converted distance to metres I ignore any units
1(f)(i)	<i>idea that:</i> no intermediate data for reaction between 7 and 8 minutes ;	1	A water
1(f)(ii)	tangent drawn <u>at 6 min</u> ; correct number in range 0.2 to 0.4 ;	2	ecf correct calculation from incorrect tangent I any sign A rate calculated from time in seconds
1(f)(iii)	curve should be to the right of original curve and must be drawn from 1–12 min ;	1	A if merges with the original after 8 min

Question	Answer	Marks	Guidance
1(g)(i)	there is no <u>significant</u> difference between the (mean) absorption values for, tubes 1 and 2 / no weed killer and weed killer X or the difference between the (mean) absorption values for, tubes 1 and 2 / no weed killer and weed killer X , is not <u>significant</u> ;	1	
1(g)(ii)	<i>1 of:</i> continuous ; normally distributed / normal distribution ; standard deviations are approximately the same ; size of the two samples is less than 30 ;	1	I 'compare the means'
1(g)(iii)	(n =) 12 ;	1	
1(g)(iv)	<u>greater than</u> (>) 2.07 / critical value or 2.07 / critical value <u>is less than</u> (<) calculated value / <i>t</i> value ;	1	